SPARKY'S SCIENCE EXPERIMENTS

Wet Cloth Cooler: Keep Water Cool Without a Fridge

Physical Science (heat) · Ages 6+ · 10 minutes, then wait 1-2 hours

Wrap one bottle of water in a wet cloth and one in a dry cloth to discover how evaporation keeps things cool - the idea behind the clay pot fridge.

Safety first

- A safe experiment. Use a shady place.

You need

- Two plastic bottles of tap water, filled at the same time
- Two pieces of cotton cloth (old socks work)
- Rubber bands or string
- A thermometer if you have one (or use your hands)

Steps

1. Wrap each bottle in cloth and hold it on with rubber bands.
2. Soak one cloth in water until it is dripping wet. Leave the other dry.
3. Stand both bottles in the shade where there is a breeze.
4. After 1 to 2 hours, feel both bottles or measure the water. Which is cooler?
5. Keep the cloth wet and try again in still air. Is the breeze important?

If it doesn't work, try these in order

1. No difference? Keep the cloth wet - it dries out quickly in wind.
2. Wait longer, or move both bottles to a breezier spot. On humid days the cooling is small.

Why it works

To evaporate (turn into an invisible gas), water needs heat. It takes that heat from the cloth, the bottle and the water inside, so they cool down. Wind carries the damp air away and speeds this up. It works best on hot, dry days; in humid weather, like summer in Durban, it cools less. Sweating cools your body the same way.

Did you know? The pot-in-pot cooler, made famous by Nigerian teacher Mohammed Bah Abba in the 1990s, uses two clay pots with wet sand between them to keep vegetables fresh without electricity.

Ask Sparky: Why do you feel cold climbing out of a pool? Would it work better in the Karoo or Durban?